# Infrastructure & Deployment Architecture

## Hardware, Network, Security & DR Design

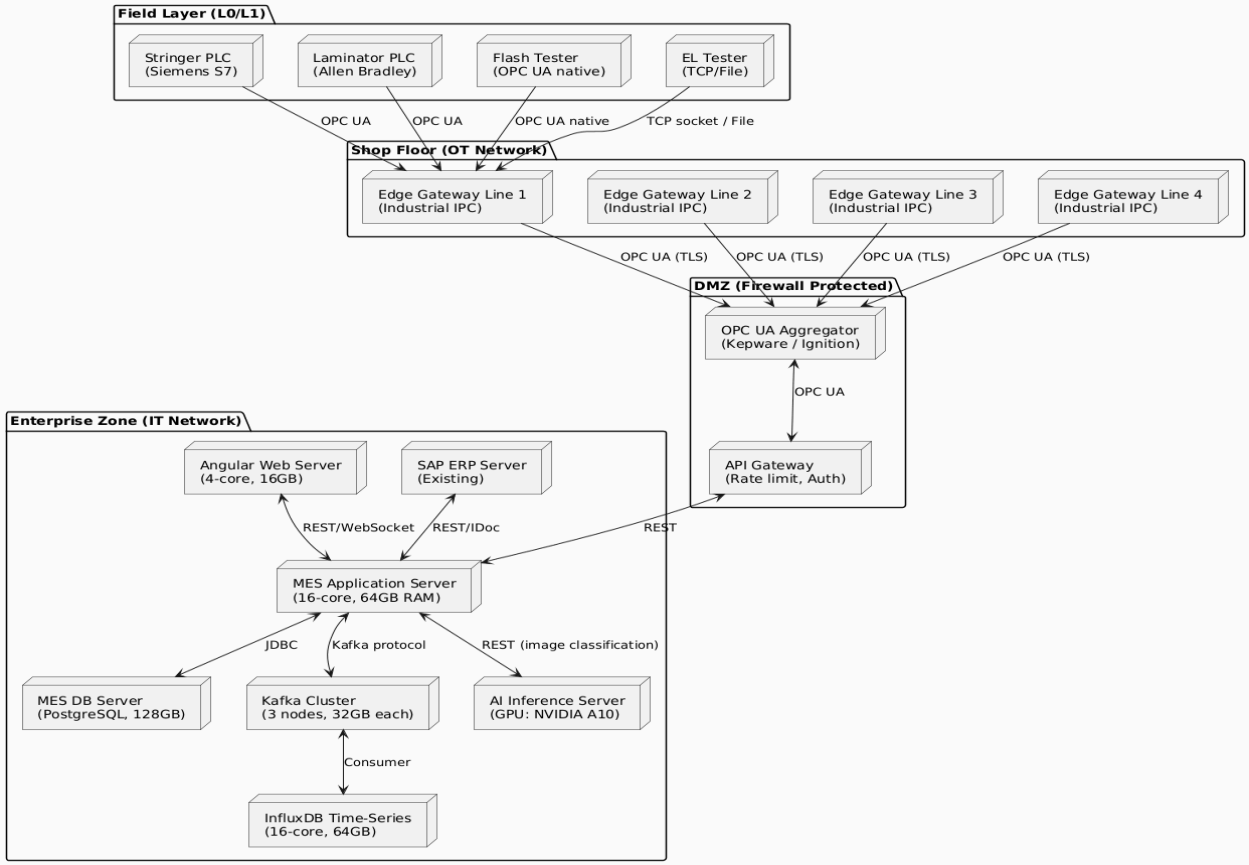
Solar Manufacturing MES — Infrastructure Blueprint | 02 May 2026

### 1. Deployment Model Overview

Hybrid deployment: Core MES, PODISE, OPC UA integration, and all real-time processing run on-premise for latency <100ms. AI inference also runs on-premise via GPU server (data sovereignty — no manufacturing IP leaves the factory). Cloud is optional for remote monitoring dashboards only.

| Layer | Component | Location | Rationale |
|---|---|---|---|
| Application | Solar MES + PODISE | On-premise | <100ms latency for shop floor operations |
| AI Inference | Ollama + Llama 3.2 + EL Classifier | On-premise GPU | Data sovereignty; no IP to cloud |
| Database | PostgreSQL (primary) | On-premise | ACID compliance; audit trail permanence |
| Time-Series | InfluxDB | On-premise | Local OEE metrics; high-write throughput |
| Messaging | Kafka (3-node cluster) | On-premise | Event replay; zero data loss |
| Edge | OPC UA Gateway per line | Shop floor | Sub-second machine data; offline buffering |
| Frontend | Angular Web Server | On-premise | Served on corporate LAN; WebSocket real-time |
| Analytics (optional) | BI / Reporting | Cloud or on-premise | Long-term trend analysis; management access |

### 2. System Deployment Diagram




## 3. Server Bill of Materials

| Server | Purpose | CPU | RAM | Storage | Qty | Est. Cost |
|---|---|---|---|---|---|---|
| MES App Server | Solar MES + PODISE + Kafka consumers | 16-core (Intel Xeon / AMD EPYC) | 64 GB ECC | 2 TB NVMe SSD (RAID-1) | 2 (HA pair) | $8,000–$12,000 each |
| DB Server (PostgreSQL) | Audit trail, genealogy, order history | 32-core | 128 GB ECC | 10 TB NVMe RAID-10 | 1 primary + 1 replica | $15,000–$20,000 each |
| InfluxDB Server | Time-series OEE, sensor metrics | 16-core | 64 GB ECC | 5 TB SSD (RAID-5) | 1 | $8,000–$12,000 |
| Kafka Broker | 3-node cluster for event streaming | 8-core each | 32 GB each | 2 TB SSD each | 3 | $5,000–$7,000 each |
| AI GPU Server | EL image classification, Ollama LLM | 16-core CPU | 128 GB RAM | 2 TB NVMe | 1 | $20,000–$35,000 |
| Web/API Server | Angular frontend, API gateway | 8-core | 32 GB | 1 TB SSD | 1 (+ HA) | $4,000–$6,000 |
| Edge IoT Gateway | OPC UA per production line (industrial IPC) | 4-core (Atom/i5) | 16 GB | 256 GB SSD (fanless) | 1 per line | $2,000–$4,000 each |

## 4. Hardware Cost Estimate Summary

| Category | Qty | Estimated Cost (USD) |
|---|---|---|
| MES Application Servers (HA pair) | 2 | $16,000–$24,000 |
| Database Server (primary + replica) | 2 | $30,000–$40,000 |
| InfluxDB Server | 1 | $8,000–$12,000 |
| Kafka Cluster (3 nodes) | 3 | $15,000–$21,000 |
| AI/GPU Inference Server | 1 | $20,000–$35,000 |
| Web/API Server | 1 | $4,000–$6,000 |
| Edge IoT Gateways (4 lines) | 4 | $12,000–$24,000 |
| OEE Display Screens (55") | 1 per line | $2400–$4000 |
| OPC UA License (Kepware/Ignition) | 1 | $10,000–$25,000 |
| Networking (switches, firewall, cables) | — | $15,000–$25,000 |
| TOTAL HARDWARE ESTIMATE | — | $92,000–$162,000 |

## 5. Network Architecture

Strict OT/IT segregation is mandatory for security. Three network zones with stateful firewalls between each zone prevent lateral movement from IT to shop floor.




| Zone | VLAN | Devices | Protocol | Firewall Rules |
|---|---|---|---|---|
| IT / Corporate LAN | VLAN 10 | SAP, MES servers, DB, Kafka, Web server | TCP/IP, HTTPS | Outbound to DMZ port 443 only; no direct OT access |
| DMZ | VLAN 20 | OPC UA aggregator, API gateway | OPC UA (4840), HTTPS | IT→DMZ: port 443; DMZ→OT: port 4840 TLS only |

| | | | | |
|---|---|---|---|---|
| OT Network | VLAN 30+ | Edge gateways, SCADA, HMIs | OPC UA, Modbus TCP | No inbound from IT; DMZ can pull on OPC UA subscription |
| Field (L0/L1) | Isolated | PLCs, instruments, vision cameras | Profibus, Ethernet/IP, OPC UA | OT network only; no internet; air-gapped from IT |

## 6. Security Architecture

| Security Layer | Requirement | Implementation |
|---|---|---|
| Authentication | MES user login; API access control | OAuth2 / JWT; LDAP/AD integration; MFA for admins |
| Authorisation | Role-based access per module | RBAC: Operator / Supervisor / Planner / Admin / Read-only |
| Data in Transit | All API, Kafka, OPC UA encrypted | TLS 1.3 mandatory; OPC UA with X.509 certificates |
| Data at Rest | PostgreSQL, InfluxDB, document store | AES-256 encryption; encrypted filesystem |
| Audit Log | Immutable action log (ALMM requirement) | Append-only audit table; 3-year retention; tamper-evident |
| OT Security | Machine whitelist; no direct internet | OPC UA authentication; MAC address filtering on OT VLAN |
| Vulnerability Mgmt | Patch management; CVE monitoring | Critical patch SLA: 48 hours; quarterly security assessment |
| Backup | Automated daily backup; off-site copy | PostgreSQL pg_dump + InfluxDB backup to encrypted NAS; weekly offsite |

## 7. High Availability & Disaster Recovery

| Requirement | Target | Implementation |
|---|---|---|
| RTO (Recovery Time Objective) | ≤ 4 hours | Hot standby MES app server; PostgreSQL streaming replication |
| RPO (Recovery Point Objective) | ≤ 1 hour | Kafka replication (3 replicas); PostgreSQL WAL archiving every 15 min |
| MES Application HA | 99.5% uptime | Active-passive HA; health check failover in <60 seconds |
| PostgreSQL HA | 99.9% DB uptime | Primary + replica; Patroni for auto-failover |
| Kafka HA | Zero message loss | 3-node cluster; replication factor 3; min-ISR 2 |
| Edge Gateway | Local buffering 4 hours | Edge stores PDI/PDO locally on connectivity loss; sync on reconnect |

| | | |
|---|---|---|
| Backup Frequency | Daily full + 15-min WAL | pg_basebackup (daily); continuous WAL archiving; InfluxDB backup daily |
| DR Drill | Quarterly test | Full failover simulation; RTO/RPO measurement; results documented |

## 8. Infrastructure Monitoring

**Application Monitoring:** Prometheus + Grafana: MES API response time, Kafka consumer lag, PostgreSQL query performance, InfluxDB write throughput

**OT Monitoring:** Edge gateway health: connectivity status per machine, OPC UA subscription latency, buffered data queue depth

**System Monitoring:** CPU, RAM, disk I/O per server; auto-alert on >80% threshold; PagerDuty / SMS escalation

**Business Monitoring:** OEE dashboard (Angular): real-time per line/machine; shift summary auto-generated by Reporting Agent

**Log Management:** Centralized logging (ELK stack or Loki); 90-day hot retention; 1-year archive; ALMM audit trail immutable